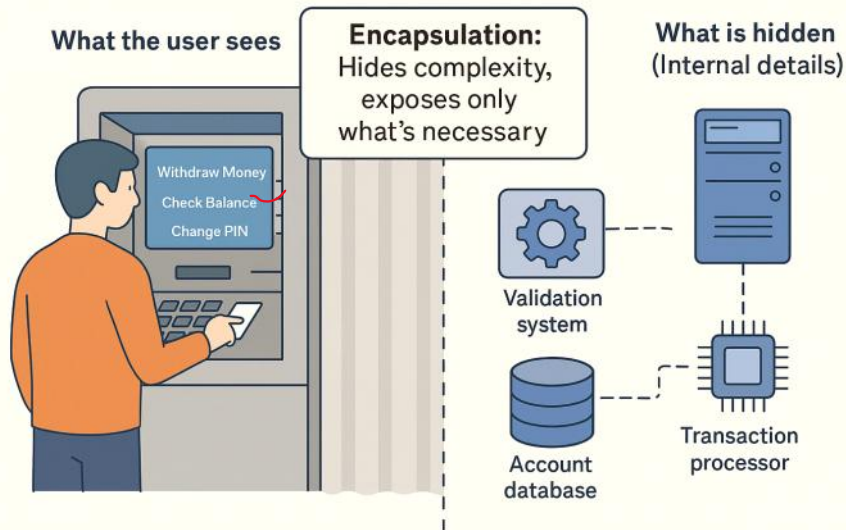


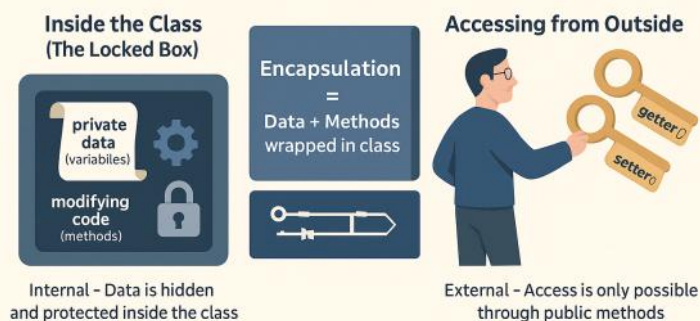
Encapsulation in Java

Encapsulation in Real Life – The ATM Example



Encapsulation in Java – The Locked Box Analogy

In Java, Encapsulation means we keep the internal data of a class safe and hidden from the outside world. We wrap the data – like variables – and the code that modifies it – like methods – inside one single unit: the class.



```
// Java Encapsulation Example
public class Student {
    // Private variables
    private String name;
    private int rollNumber;

    // Getter for name
    public String getName() {
        return name;
    }

    // Setter for name
    public void setName(String name) {
        // You can add validations here
        if (name != null && !name.isEmpty()) {
            this.name = name;
        }
    }

    // Getter for rollNumber
    public int getRollNumber() {
        return rollNumber;
    }

    // Setter for rollNumber
    public void setRollNumber(int rollNumber) {
        // Example validation
        if (rollNumber > 0) {
            this.rollNumber = rollNumber;
        }
    }
}
```

Encapsulation in Real Life – The Coffee Machine Example

